

The fourth-degree polynomial

$$f(x) = 230x^4 + 18x^3 + 9x^2 - 221x - 9$$

has two real zeros, one in $[-1, 0]$ and the other in $[0, 1]$. Attempt to approximate these zeros to within 10^{-6} using the

- a. Method of False Position
- b. Secant method
- c. Newton's method

Use the endpoints of each interval as the initial approximations in (a) and (b) and the midpoints as the initial approximation in (c).

(a) False Position Method

Upgrade

The formula is

$$p_n = \frac{af(b) - bf(a)}{f(b) - f(a)}.$$

Starting with

$$a = -1, \quad b = 0.$$

The iterations are:

n	a_n	b_n	p_n	$f(p_n)$
1	-1	0	-0.020361991	-4.496381
2	-1	-0.020362	-0.030430247	-2.266891
3	-1	-0.030430	-0.035479814	-1.148071

(b) Secant Method

The Secant formula is

$$p_n = p_{n-1} - f(p_{n-1}) \frac{p_{n-1} - p_{n-2}}{f(p_{n-1}) - f(p_{n-2})}.$$

Use

$$p_0 = -1, \quad p_1 = 0.$$

Iterations

n	p_{n-1}	p_n	$f(p_n)$
2	-1	-0.020361991	-4.496381
3	0	-0.040691256	7.087×10^{-3}
4	-0.020361991	-0.040659263	-5.706×10^{-6}

(c) Newton's Method

Newton's formula is

$$p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}.$$

The midpoint of $[-1, 0]$ is

$$p_0 = \frac{-1 + 0}{2} = -0.5.$$

Therefore,

$$p_0 = -0.5.$$

Iterations

n	p_n	p_{n+1}	$f(p_{n+1})$
0	-0.500000000	-0.150452489	24.510271
1	-0.150452489	-0.041816814	0.256641
2	-0.041816814	-0.040659343	1.2234×10^{-5}
3	-0.040659343	-0.040659288	2.84×10^{-14}

(a) False Position Method

Use

$$a = 0, \quad b = 1.$$

The iterations are:

n	a_n	b_n	p_n	$f(p_n)$
1	0	1	0.250000000	-62.507813
2	0.25	1	0.773762765	-83.830520
3	0.773763	1	0.944885169	-11.265131

(b) Secant Method

Now use the endpoints

$$p_0 = 0, \quad p_1 = 1.$$

Here something interesting happens.

The iterations are:

n	p_{n-1}	p_n	$f(p_n)$
2	0	1	-62.507813
3	1	0.250000	-83.830520
4	0.250000	-1.285418	879.638986

(c) Newton's Method

The required initial approximation is the midpoint:

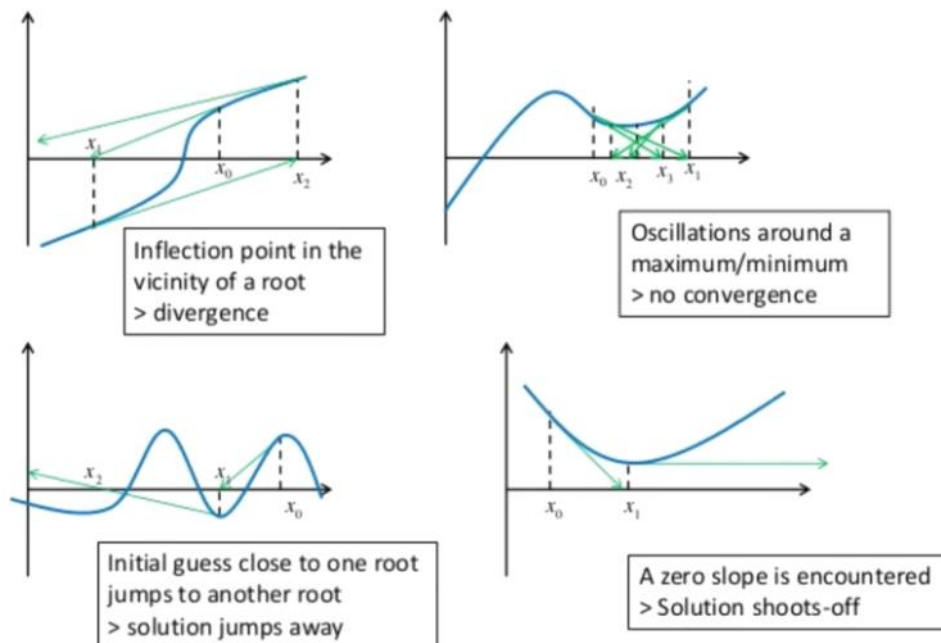
$$p_0 = \frac{0 + 1}{2} = 0.5.$$

Use

$$p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}.$$

The iterations are:

n	p_n	p_{n+1}	$f(p_{n+1})$
0	0.500000000	-0.705089820	201.836304
1	-0.705089820	-0.323791114	65.418426
2	-0.323791114	-0.064603131	5.314007



Multiple Roots

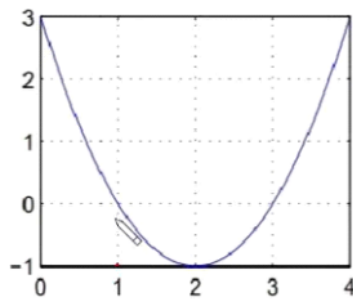
For a function $f(x) = 0$.

The root ξ is called **multiple roots** of $f(x)$
with **multiplicity** m when

$$f(\xi) = 0 ; f'(\xi) = 0 ; \dots f^{m-1}(\xi) = 0 ;$$

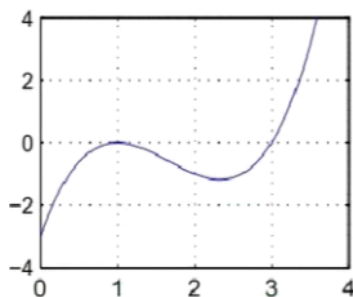
$$f^m(\xi) \neq 0$$

$$f(x) = (x - 1)(x - 3)$$



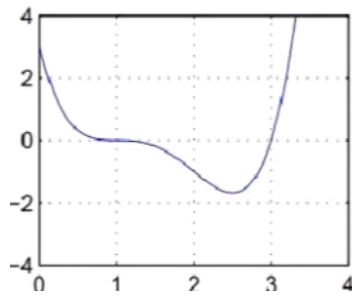
$$f(x) = (x - 1)^2 (x - 3)$$

$$x_{n+1} = x_n - \frac{f(x)}{f'(x)}$$



- The **sign of the function** does not change around **even multiple roots**
- Both $f(x)$ and $f'(x)$ go to zero around multiple roots
- Newton-Raphson and secant methods **may converge slowly** or even diverge.

$$f(x) = (x - 1)^3 (x - 3)$$



Thus, various methods such as

- Bisection Method
- Regula-Falsi method
- Secant Method
- Fixed-point iteration method
- Newton-Raphson method

If ξ is root of $f(x) = 0$ with multiplicity m ,

Then the iterative formula for multiple roots becomes

$$x_{n+1} = x_n - m \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = 0.5 - 2 \left(\frac{\quad}{\quad} \right)'$$

Example: Find the double root of the equation

$$f(x) = x^3 - x^2 - x + 1 = 0$$

$$x_0 = 0.5$$

Case 2: When multiplicity of $f(x)$ not given

$$x_{n+1} = x_n - \frac{f(x_n)f'(x_n)}{[f'(x_n)]^2 - f(x_n)f''(x_n)}$$

Example: For the equation

$$f(x) = x^3 - 7x^2 + 16x - 12 = 0,$$

find the root correct to three decimal places using Modified newton's method with initial approximation $x = 1$.

Example:

Find the multiplicity of the root $x = 1$
for equation $(x - 1)^2 \ln x = 0$.

Apply modified Newton's method with
initial approximation 0.5 and
perform three iterations.

Solution:

By modified Newton's method

$$\begin{aligned}x_{n+1} &= x_n - 3 \frac{f(x_n)}{f'(x_n)} \\&= \frac{2x_n \ln x_n + (x_n - 1)[1 - 3 \ln x_n]}{2 \ln x_n + \frac{(x_n - 1)}{x_n}}\end{aligned}$$

Example:

If $f(x) = e^x - 1 - x$. Check whether
 f has a zero of multiplicity 2 at $x = 0$.

If yes, apply modified Newton's Raphson
method with $x_0 = 1$ to find correct root
at four decimal places.

Order of Convergence

Definition 2.7 Suppose $\{p_n\}_{n=0}^{\infty}$ is a sequence that converges to p , with $p_n \neq p$ for all n . If positive constants λ and α exist with

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|^\alpha} = \lambda,$$

then $\{p_n\}_{n=0}^{\infty}$ converges to p of order α , with asymptotic error constant λ . ■

An iterative technique of the form $p_n = g(p_{n-1})$ is said to be of order α if the sequence $\{p_n\}_{n=0}^{\infty}$ converges to the solution $p = g(p)$ of order α .

In general, a sequence with a high order of convergence converges more rapidly than a sequence with a lower order. The asymptotic constant affects the speed of convergence but not to the extent of the order. Two cases of order are given special attention.

- (i) If $\alpha = 1$ (and $\lambda < 1$), the sequence is **linearly convergent**.
- (ii) If $\alpha = 2$, the sequence is **quadratically convergent**.

Illustration Suppose that $\{p_n\}_{n=0}^{\infty}$ is linearly convergent to 0 with

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1}|}{|p_n|} = 0.5$$

and that $\{\tilde{p}_n\}_{n=0}^{\infty}$ is quadratically convergent to 0 with the same asymptotic error constant,

$$\lim_{n \rightarrow \infty} \frac{|\tilde{p}_{n+1}|}{|\tilde{p}_n|^2} = 0.5.$$

For simplicity we assume that for each n we have

$$\frac{|p_{n+1}|}{|p_n|} \approx 0.5 \quad \text{and} \quad \frac{|\tilde{p}_{n+1}|}{|\tilde{p}_n|^2} \approx 0.5.$$

For the linearly convergent scheme, this means that

$$|p_n - 0| = |p_n| \approx 0.5|p_{n-1}| \approx (0.5)^2|p_{n-2}| \approx \dots \approx (0.5)^n|p_0|,$$

whereas the quadratically convergent procedure has

$$\begin{aligned} |\tilde{p}_n - 0| = |\tilde{p}_n| &\approx 0.5|\tilde{p}_{n-1}|^2 \approx (0.5)[0.5|\tilde{p}_{n-2}|^2]^2 = (0.5)^3|\tilde{p}_{n-2}|^4 \\ &\approx (0.5)^3[(0.5)|\tilde{p}_{n-3}|^2]^4 = (0.5)^7|\tilde{p}_{n-3}|^8 \\ &\approx \dots \approx (0.5)^{2^n-1}|\tilde{p}_0|^{2^n}. \end{aligned}$$

Table 2.7

n	Linear Convergence Sequence $\{p_n\}_{n=0}^{\infty}$ $(0.5)^n$	Quadratic Convergence Sequence $\{\tilde{p}_n\}_{n=0}^{\infty}$ $(0.5)^{2^n-1}$
1	5.0000×10^{-1}	5.0000×10^{-1}
2	2.5000×10^{-1}	1.2500×10^{-1}
3	1.2500×10^{-1}	7.8125×10^{-3}
4	6.2500×10^{-2}	3.0518×10^{-5}
5	3.1250×10^{-2}	4.6566×10^{-10}
6	1.5625×10^{-2}	1.0842×10^{-19}
7	7.8125×10^{-3}	5.8775×10^{-39}

Theorem 2.8 Let $g \in C[a, b]$ be such that $g(x) \in [a, b]$, for all $x \in [a, b]$. Suppose, in addition, that g' is continuous on (a, b) and a positive constant $k < 1$ exists with

$$|g'(x)| \leq k, \quad \text{for all } x \in (a, b).$$

If $g'(p) \neq 0$, then for any number $p_0 \neq p$ in $[a, b]$, the sequence

$$p_n = g(p_{n-1}), \quad \text{for } n \geq 1,$$

converges only linearly to the unique fixed point p in $[a, b]$. ■

Proof We know from the Fixed-Point Theorem 2.4 in Section 2.2 that the sequence converges to p . Since g' exists on (a, b) , we can apply the Mean Value Theorem to g to show that for any n ,

$$p_{n+1} - p = g(p_n) - g(p) = g'(\xi_n)(p_n - p),$$

where ξ_n is between p_n and p . Since $\{p_n\}_{n=0}^{\infty}$ converges to p , we also have $\{\xi_n\}_{n=0}^{\infty}$ converging to p . Since g' is continuous on (a, b) , we have

$$\lim_{n \rightarrow \infty} g'(\xi_n) = g'(p).$$

Thus

$$\lim_{n \rightarrow \infty} \frac{p_{n+1} - p}{p_n - p} = \lim_{n \rightarrow \infty} g'(\xi_n) = g'(p) \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|} = |g'(p)|.$$

Hence, if $g'(p) \neq 0$, fixed-point iteration exhibits linear convergence with asymptotic error constant $|g'(p)|$. ■ ■ ■

Theorem 2.9 Let p be a solution of the equation $x = g(x)$. Suppose that $g'(p) = 0$ and g'' is continuous with $|g''(x)| < M$ on an open interval I containing p . Then there exists a $\delta > 0$ such that, for $p_0 \in [p - \delta, p + \delta]$, the sequence defined by $p_n = g(p_{n-1})$, when $n \geq 1$, converges at least quadratically to p . Moreover, for sufficiently large values of n ,

$$|p_{n+1} - p| < \frac{M}{2} |p_n - p|^2. \quad \blacksquare$$

Proof Choose k in $(0, 1)$ and $\delta > 0$ such that on the interval $[p - \delta, p + \delta]$, contained in I , we have $|g'(x)| \leq k$ and g'' continuous. Since $|g'(x)| \leq k < 1$, the argument used in the proof of Theorem 2.6 in Section 2.3 shows that the terms of the sequence $\{p_n\}_{n=0}^{\infty}$ are contained in $[p - \delta, p + \delta]$. Expanding $g(x)$ in a linear Taylor polynomial for $x \in [p - \delta, p + \delta]$ gives

$$g(x) = g(p) + g'(p)(x - p) + \frac{g''(\xi)}{2}(x - p)^2,$$

where ξ lies between x and p . The hypotheses $g(p) = p$ and $g'(p) = 0$ imply that

$$g(x) = p + \frac{g''(\xi)}{2}(x - p)^2.$$

In particular, when $x = p_n$,

$$p_{n+1} = g(p_n) = p + \frac{g''(\xi_n)}{2}(p_n - p)^2,$$

with ξ_n between p_n and p . Thus,

$$p_{n+1} - p = \frac{g''(\xi_n)}{2}(p_n - p)^2.$$

Since $|g'(x)| \leq k < 1$ on $[p - \delta, p + \delta]$ and g maps $[p - \delta, p + \delta]$ into itself, it follows from the Fixed-Point Theorem that $\{p_n\}_{n=0}^{\infty}$ converges to p . But ξ_n is between p and p_n for each n , so $\{\xi_n\}_{n=0}^{\infty}$ also converges to p , and

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|^2} = \frac{|g''(p)|}{2}.$$

Multiple Roots

Multiple Roots

Definition 2.10 A solution p of $f(x) = 0$ is a **zero of multiplicity m** of f if for $x \neq p$, we can write $f(x) = (x - p)^m q(x)$, where $\lim_{x \rightarrow p} q(x) \neq 0$. ■

Theorem 2.12 The function $f \in C^m[a, b]$ has a zero of multiplicity m at p in (a, b) if and only if

$$0 = f(p) = f'(p) = f''(p) = \cdots = f^{(m-1)}(p), \quad \text{but } f^{(m)}(p) \neq 0. \quad \blacksquare$$

Example 1 Let $f(x) = e^x - x - 1$. (a) Show that f has a zero of multiplicity 2 at $x = 0$. (b) Show that Newton's method with $p_0 = 1$ converges to this zero but not quadratically.

Table 2.8

n	p_n
0	1.0
1	0.58198
2	0.31906
3	0.16800
4	0.08635
5	0.04380
6	0.02206
7	0.01107
8	0.005545
9	2.7750×10^{-3}
10	1.3881×10^{-3}
11	6.9411×10^{-4}
12	3.4703×10^{-4}
13	1.7416×10^{-4}
14	8.8041×10^{-5}

Solution (a) We have

$$f(x) = e^x - x - 1, \quad f'(x) = e^x - 1 \quad \text{and} \quad f''(x) = e^x,$$

so

$$f(0) = e^0 - 0 - 1 = 0, \quad f'(0) = e^0 - 1 = 0 \quad \text{and} \quad f''(0) = e^0 = 1.$$

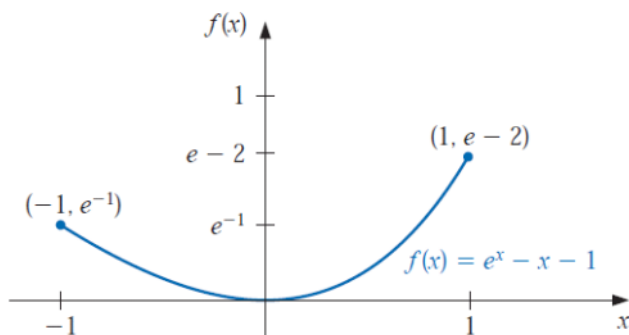
Theorem 2.12 implies that f has a zero of multiplicity 2 at $x = 0$.

(b) The first two terms generated by Newton's method applied to f with $p_0 = 1$ are

$$p_1 = p_0 - \frac{f(p_0)}{f'(p_0)} = 1 - \frac{e - 2}{e - 1} \approx 0.58198,$$

and

$$p_2 = p_1 - \frac{f(p_1)}{f'(p_1)} \approx 0.58198 - \frac{0.20760}{0.78957} \approx 0.31906.$$



Example 2 In Example 1 it was shown that $f(x) = e^x - x - 1$ has a zero of multiplicity 2 at $x = 0$ and that Newton's method with $p_0 = 1$ converges to this zero but not quadratically. Show that the modification of Newton's method as given in Eq. (2.13) improves the rate of convergence.

Solution Modified Newton's method gives

$$p_1 = p_0 - \frac{f(p_0)f'(p_0)}{f'(p_0)^2 - f(p_0)f''(p_0)} = 1 - \frac{(e - 2)(e - 1)}{(e - 1)^2 - (e - 2)e} \approx -2.3421061 \times 10^{-1}.$$

This is considerably closer to 0 than the first term using Newton's method, which was 0.58918. Table 2.9 lists the first five approximations to the double zero at $x = 0$. The results were obtained using a system with ten digits of precision. The relative lack of improvement in the last two entries is due to the fact that using this system both the numerator and the denominator approach 0. Consequently there is a loss of significant digits of accuracy as the approximations approach 0. ■

Table 2.9

n	p_n
1	$-2.3421061 \times 10^{-1}$
2	$-8.4582788 \times 10^{-3}$
3	$-1.1889524 \times 10^{-5}$
4	$-6.8638230 \times 10^{-6}$
5	$-2.8085217 \times 10^{-7}$